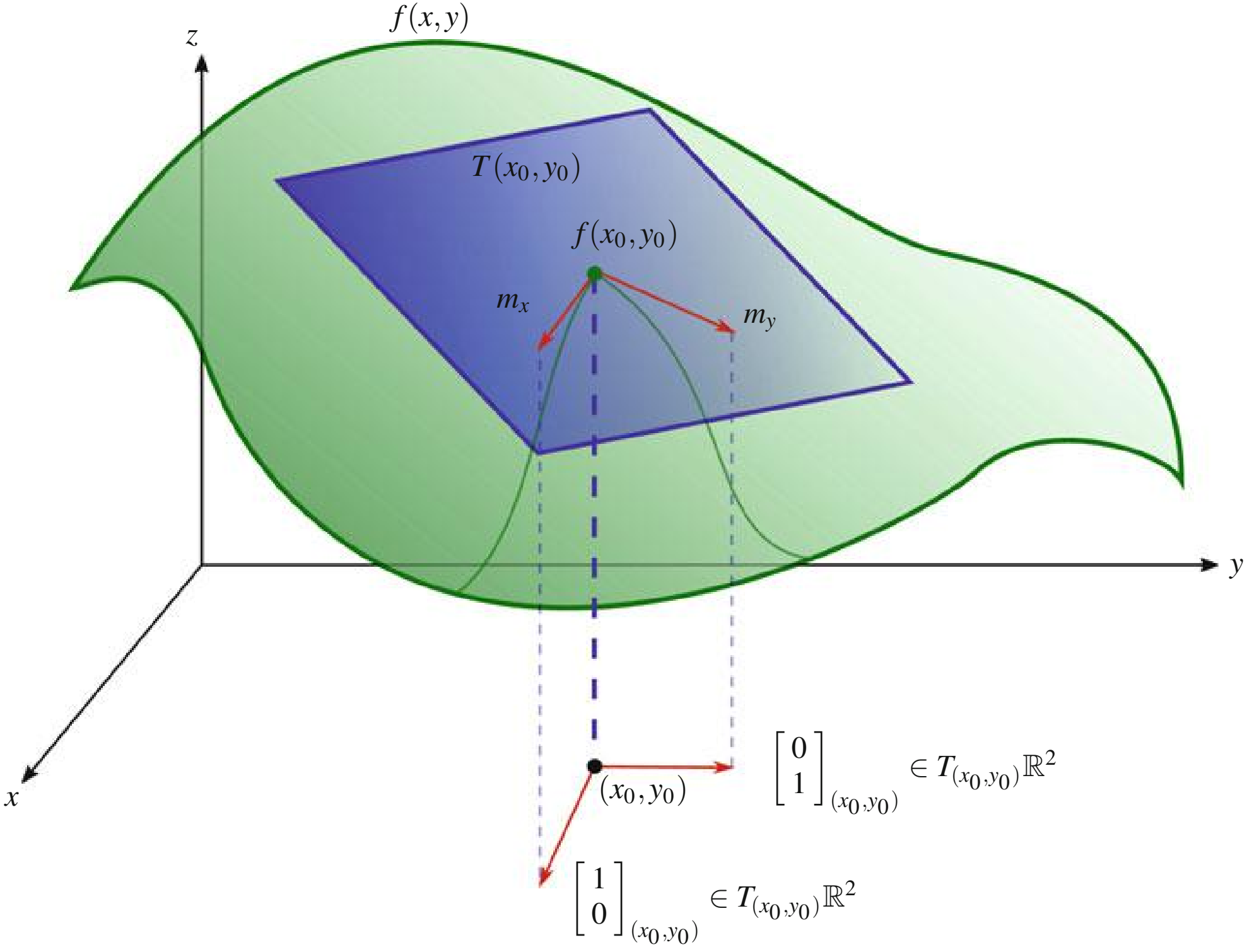

$f(x,y)$
z
$T(x_0,y_0)$
$f(x_0,y_0)$
m_x
m_y
y
x
(x_0,y_0)
$\begin{bmatrix}0\\1\end{bmatrix}_{(x_0,y_0)} \in T_{(x_0,y_0)}\mathbb{R}^2$
$\begin{bmatrix}1\\0\end{bmatrix}_{(x_0,y_0)} \in T_{(x_0,y_0)}\mathbb{R}^2$